

TECHNICAL BULLETIN

DIXIE CHEMICAL
10601 Bay Area Blvd.
Pasadena, TX 77507
Tel: (281) 474-3271
Fax: (281) 291-3384
E-Mail: info@dixiechemical.com

**EPOXY CURING AGENT 150
(ECA 150)****Typical Properties**

Property	Value	Test Method
Molecular Weight (average)	163	
Specific gravity @ 25°C, g/cc	1.190 - 1.210	D-102 (ASTM D 4052)
Appearance	Clear to light yellow liquid	D-104 (Visual)
Viscosity (Brookfield at 25 °C), cP	40 - 70	D-150F
Color, Gardner	2 Max	D-150G
MTHPA+MHHPA	83.0-86.0	D-160
THPA+HHPA	14.0-17.0	D-160
Free acid, %	1 Max	D 160B-2
Volatiles, ppm	100 Max	D-160A
Maleic Anhydride, %	0.1 Max	D-160

Applications

ECA 150 is a light colored, low viscosity liquid epoxy curing agent based on methyltetrahydrophthalic anhydride. ECA 150 is designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. This product performs well in demanding applications such as filament winding, pultrusion and other forms of lamination. It is also used extensively in electrical encapsulation, insulation, casting and potting.

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical Company, Inc. makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

TECHNICAL BULLETIN

ECA 150

ECA 150 imparts ease of handling due to low viscosity, low toxicity and low volatility. It is a liquid at normal room temperature. If the product crystallizes, it can be returned to liquid by heating 38-55°C (100-130°F) for 2-8 hours.

For more information on the use of anhydrides like ECA 150 as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical.

Formulation Example

ECA 150 was formulated with a variety of epoxy resins using 1 phr of AMI-1 as a catalyst. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 150, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	85	840	39	151
Low Viscosity BPA Liquid	88	600	37	151
Cycloaliphatic Epoxy	117	134	40	228
Epoxy Phenol Novolac	93	790	35	146
Epoxy BPA Novolac	84	2860	36	160
BPF Liquid Epoxy	95	400	38	134

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical Company, Inc. makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

TECHNICAL BULLETIN

ECA 150

ECA 150 will react with water to form diacids. This is normally undesirable, so ECA 150 should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for ECA 150, consult the Dixie Chemical brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 150 is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.